

ISHA ADHIKARI

Junior AI Engineer | MSc Data Science Candidate

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NEC Registered Computer Engineer | License No. 90960

PROFESSIONAL SUMMARY

Computer Engineering graduate and MSc Data Science candidate focused on building reliable AI-agent systems in Python. Recently built and deployed an LLM agent with LangGraph and FastAPI whose design centres on reliability — structured (Pydantic-validated) outputs, a self-grading retrieve-retry loop with fallback paths, PostgreSQL persistence, Docker packaging, and pytest-backed GitHub Actions CI. Prior professional experience as a Java developer working with PostgreSQL and REST APIs. Seeking a Junior AI Engineer role building production agent workflows.

TECHNICAL SKILLS

Languages: Python, Java, R, SQL

LLMs & Agents: LangGraph, agent workflows, tool/function calling, LLM APIs, prompt engineering & iteration, RAG, retry/fallback logic

Backend & APIs: FastAPI, REST APIs, PostgreSQL, Pydantic (structured-output validation)

Reliability & Testing: pytest, GitHub Actions CI, regression testing, Docker

ML / Deep Learning: scikit-learn, TensorFlow/Keras, XGBoost, transfer learning (MobileNet), TFLite

Big Data & NLP: PySpark, Spark MLlib; TF-IDF, cosine-similarity retrieval, Naive Bayes; pandas, NumPy

PROJECTS

Scholar Agent — Reliable LLM Agent API [\[github.com/Isha-12345/scholar-agent \]](https://github.com/Isha-12345/scholar-agent)

Python · LangGraph · FastAPI · PostgreSQL · Pydantic · Docker · pytest · GitHub Actions

- Designed a LangGraph agent around reliability: it routes each query, retrieves from a PostgreSQL store, grades whether its own results are relevant, and rewrites-and-retries on failure with a bounded fallback path — a self-correcting agent, not a single-shot RAG chatbot.
- Enforced structured outputs with Pydantic schemas, exposed the agent as a FastAPI REST service, containerized it with Docker Compose, and covered it with pytest tests run in a GitHub Actions CI pipeline.

Intelligent Security & Access Control System [\[github.com/Isha-12345/ann-face-access-control \]](https://github.com/Isha-12345/ann-face-access-control)

Python · TensorFlow/Keras · MobileNet · TFLite · Raspberry Pi

- Three-class access control (known / unknown / intrusion) with MobileNet transfer learning and open-set anomaly detection; quantized to INT8 TFLite and deployed on a Raspberry Pi with GPIO alerts.

Flight Delay Analysis at Scale [\[github.com/Isha-12345/flight-delay-bigdata \]](https://github.com/Isha-12345/flight-delay-bigdata)

PySpark · Spark MLlib · Tableau

- Distributed PySpark pipeline over a large flights dataset — classification, regression, and K-Means clustering with Spark MLlib — with results exported to Tableau dashboards.

CU Research Portal & Data-Center PUE Prediction [\[github.com/Isha-12345 \]](https://github.com/Isha-12345)

Python · Selenium · TF-IDF · Naive Bayes · scikit-learn · XGBoost

- Academic search engine (custom TF-IDF index + cosine ranking + Naive Bayes classifier); and a Random Forest / XGBoost / SVR study predicting data-center PUE and CO₂ emissions with cross-validation.

EXPERIENCE

Computer Science Instructor — National School of Sciences (NIST), Lainchaur, Kathmandu 2025 – Present

- Teach computer-science subjects to secondary-level students, explaining technical concepts clearly and in structured form.

Freelance Web Developer — Self-employed

2023 – Present

- Design and build responsive websites for clients, handling front-end implementation and delivery.

Junior Java Developer — Phoenix Solutions

Sep 2024 – Feb 2025

- Developed and maintained Java application features backed by PostgreSQL and REST APIs, using Git-based workflows.

Java Developer Intern — Phoenix Solutions

Jun 2024 – Sep 2024

- Contributed to Java development tasks and gained experience with relational databases and team workflows.

EDUCATION

MSc Data Science — Coventry University (Softwarica College, Kathmandu)

2025 – Present

BE Computer Engineering — Sagarmatha Engineering College

2019 – 2024

Higher Secondary (+2), Science — Kathmandu Model Higher Secondary School

2017 – 2019